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Svenja Keune
University of Borås

Åsa Ståhl
Linneaus University

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LEARNING FROM SEASONAL INTERVENTIONS: A SEASONAL DESIGNING FRAMEWORK

SVENJA KEUNE
UNIVERSITY OF BORÅS
SVENJA.KEUNE@HB.SE

ÅSA STÅHL
LINNAEUS UNIVERSITY
ASA.STAHL@LNU.SE

ABSTRACT

Imagine it is a rainy winter day, and you are walking up to a little mustard yellow house on wheels and step up three wooden steps to enter. Some people are sitting around the table. You take a step inside to join and have some tea. While stepping in you hear the splash of a puddle and feel your socks soak up water before you even get your foot back on the doorstep. What just happened? We built the Holding Surplus House to explore relations to resources and today's challenges through a series of seasonal experimentations. These interventions enable physical experiences as a starting point to become aware and engage in discussion about the underlying themes of householding with resources in a holistic way. In this exploratory article we look back at a full cycle of experiments and do so by reflecting on the seasons they were situated in. As a result, we suggest designers to work with relationships between temporality, place and seasonal forces into constantly emerging designs. Furthermore, we introduce the seasonal designing framework as a practical and conceptual response to contemporary grand and complex eco-social challenges and life-long learning.

INTRODUCTION

The Holding Surplus House (HSH) investigates the role and potential of households as a starting point for relating to the built environment and today's challenges that are deeply entangled with how we handle resources in our domestic homes just as well as our planetary home. Grounded in permaculture principles—care for the earth, care for people, and fair share of surplus—the HSH serves as both a physical and conceptual framework for investigating transformative practices and imaginaries of alternative ways of living.

HOUSEHOLDING

The physical framework consists of a Tiny House on Wheels (THoW), a lightweight construction on a boat trailer that can be moved around without too much effort. The THoW allows us to explore alternative ways of living (Keune, 2019). In this paper we refer to the THoW as well as the Holding Surplus House project as HSH. Rather than being richly equipped to generate and share surplus, the mobile house challenges conventional notions of resources—how they are acquired, organized, valued, and fairly distributed. It is dependent on rethinking what resources are and that most, if not all of them, are of relational nature. Rather than presenting the house as a finished entity or a model for self-sufficient off-grid living, HSH foregrounds relationality and ongoing processes of becoming —emphasizing collaboration, vulnerability, and incompleteness (Ståhl et al., 2025). A relationship-building activity that comes along with the HSH is to simply ask for help, admitting and recognising that it is an entity that needs other actors, materials and processes. We invite for collective experiences, brainstorming and reflection on a regular basis, advancing our understanding of the HSH through the participation of others (Cerna et al., 2024; Mugnaini et al., 2025).

The conceptual framework builds on the greek term oikos that relates to providing for those that share a house, or householding. Not only does ecology and

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economy meet in the oikos, literally through etymological origin, but also practically.

Bittner (2015), a cultural theorist and expert on the Bauhaus legacy that we live in, particularly in the Western world, writes about oikos: “Reflecting on the household also means addressing fundamental questions about the human constitution. In general, Western humanism, which sees man as the counterplan to nature, awarded great importance to the home and to housekeeping. The taming of a nature perceived as external, the cultivation of land and settlement were regarded as central anthropological narratives. The description of domesticated communities thereby focused on the household as a dynamic unit which in itself represents a cultural order and in which production and economic reproduction, the storage of resources, bequeathing and distributing them took place.”

Economy for us is thus not only that which is monetary. As a premise to understand economy Gibson-Graham and Miller (2015, p.8) ask: “What if we were to see the economy as [italics in original text] *ecology* – as a web of human ecological behaviors no longer bounded but fully integrated into a complex flow of ethical and energetic interdependencies; births, contaminations, self-organizings, mergings, extinctions, and patterns of habitat maintenance and destruction?“

The word itself, economy, has its roots in oikos and means habitat, whereas nomos means negotiation of order (ibid, p.12). Ecology also has its roots in oikos, but the ending “logos” means that it encompasses “account of” habitat. The authors urge us to identify “key sites of ethical negotiation” and that “an economic ethics for the Anthropocene calls us to become practiced in negotiating” participation, necessity or sufficiency, surplus and commons. Economy thus stems from the habitat, and negotiating or thinking is integral when taking economy *as* ecology. Creating gatherings for negotiation in things is an integral part of recent design history (Binder et al., 2011) expressed for example as designing in relation to ongoing living with technologies in temporary assemblies (Lindström & Ståhl, 2012; 2014a; 2014b). Design thus has a legacy of mundane negotiations although it is not often brought to the topic of everyday economic and ecological decisions such as householding with resources in households amongst different householders. Putting oneself in relationships by asking for help, as mentioned earlier, is one way of negotiating resource usage and of making economy and ecology at the same time.

In the chapter “The World as one Household”, Gibson (2015, p.316) claims that “We need to attend to the specificity and geography of habitats and can do most if we start where we are, in our own ‘high energy, high consumption, and hyper-instrumental societies’”. At the same time, the title of the chapter indicates that we are not only connected to where we set foot. We share the world as a household. Entanglements of how to live well and questions about the built environment in which we go about with our everyday lives and make different marks on the planet are thus not only a matter for one household, nor only for a local community, although it always starts there.

WORKING WITH THE SEASONS

Modernism has taught and brought us to strive for universal systems that operate detached from the resources of the local place and seasons. Modern buildings answer to standardised and climate technologies, disregarding the local building traditions, materials, resources and traditional knowledge. This universalism and the patriarchal design approaches in which anything should be possible and everything always available, has highly influenced design environments and approaches. With the rising awareness of the pressing environmental challenges and the turn of design towards more-than-human perspectives, locality and seasonality turn back to the design practice. The potential of seasons as a resource to guide the design process and its outcomes however, hasn´t received much articulated attention.

This explorative article therefore introduces the Seasonal Designing Framework and exemplifies the process of seasonal designing through a series of seasonal interventions.

SEASONAL INTERVENTIONS

Since 2021 we, the HSH project team that apart from the two co-authors also consist at its core of Stephan Hruza, Mathilda Tham and Växjö Art Gallery, have supported the house in building relations to different communities and projects. This entails inviting others to join or to use our framework for their ideas. One of our own series of explorations became seasonal interventions, guided by situated interests and context that have changed over time and that have been building up as responses to what came before.

We define seasonal interventions as artistic installations that relate to place and seasonality by e.g. using temperature, wind, water, sun, and natural materials in their seasonal state, e.g. fresh leaves in summer and dry leaves in autumn, as resources for the installation. The seasonal interventions are furthermore categorised by being part of a personal and/or collective investigation that includes the physical and conceptual experience of the installation by the artist/author and/or the audience/participants.

As a contribution to design students and practitioners, this paper leads us to encourage those conducting design projects to a) align their design projects to the seasons as an easy starting point to relate to the environment/local context and the more-than-human

and b) to stage interventions that allow themselves and others to experience and explore an environmental challenge in a playful way to encourage creative and innovative everyday practices that can scale out from that particular situation (Mugnaini et al., 2025).

GROWING A WINDOW BLIND: SEEDING RITUAL (SPRING 2024)



Figure 1: The floor covered in a mix of seeds and soil. Svenja Keune, 2024

Spiralling out from the need to for a window blind to keep out sun and curious eyes from passersby, we were inspired by the works of Scherer (Zhou et al., 2021) and Holloway (2022) we materialised the idea of making it possible to experience a floor of grass. We set out to harness the temperature and light in spring 2024 to grow a grassy window blind inside the HSH. As a starting point, we chose the opening of the Garden Futures exhibition at the art and design museum Vandalorum in Värnamo, Sweden. We covered the wooden floor of the HSH with a sheet of plastic to prevent it from soaking water and drove for a good hour from Växjö to Värnamo. There, we engaged participants of the "Sun in my backyard" (Ståhl et al., 2024) workshop that took place on the day of the opening in a seeding ritual. Having prepared a mix of planting soil and barley grass seeds, we invited the participants to take a handful, spend a moment to feel and smell the soil, and carry the mix as well as an intention into the house, covering the floor one handful at a time. The atmosphere was light and happy, inviting laughter and even an improvised version of a Swedish cattle-calling song (Här är den du letar efter) into the experience. The following weeks, the house and seeds were cared for by the museum's staff and dedicated gardener. The growth was noticeable for the visitors, although we did not invite anybody to step on the floor at this time.

GROWING A WINDOW BLIND: HARVEST (SUMMER 2024)



Figure 2: Juicing the barley grass (left). The root textile in process besides drinks with barley grass ice cubes. Åsa Ståhl (left), Svenja Keune (right), 2024

A couple of weeks before the Southern Sweden Design Days (SSDD) during which we wanted to exhibit the grown window blind, we transported the house back to the Linnaeus University Campus in Växjö to harvest and prepare the root system. We split the grassy carpet into pieces the size of our imagined window blind and cut the grass with scissors. Since we wanted to generate an experience for ourselves and the visitors that was as immersive and experiential as possible and since barley grass is highly nutritious and served as expensive shots in juice bars, we juiced the barley grass and froze them into ice cubes. Separating the roots from the soil prove very difficult and the network of roots was quite thin and flimsy in several places. This led us to sharing this part of the process with the audience of the SSDD. Therefore, we invited students and colleagues to co-create formats through which we could explore the potential of this project together.

MENDING WHAT WAS NEVER BROKEN AND DRINKING THE REST (SUMMER 2024)



Figure 3: Mending the root textile with a thick jute thread during the seminar. Svenja Keune, 2024

One of those collaborations was the "Mending what was never broken" seminar with Hanna Hofverberg and Donna Maione. In what we called an un-seminar, we

gathered at the SSDD to engage with the root textile through the textile practice of mending. The engagement led to a set of questions/inquiries about how to mend something that was, from the perspective of the grass, never broken and how to engage with a root system, that has its own logic for how to interweave itself with needle and thread. Throughout the SSDD we engaged the audience in our questions and besides serving a barley grass juice drink to about a hundred people we shared all the different stages of growing a window blind inside the HSH. We kept a section of the grass floor growing in the THoW for visitors to experience what it was like to share the indoors with this living matter. Some reacted by not wanting to step on it since it looked so vibrant and delicate. They climbed on the benches along the wall. Others wanted to touch it – and a dog wanted to eat it.

LIVING WITH A FOREST FLOOR (AUTUMN 2024)

The autumn intervention was motivated by the struggle of animals in winter to find suitable habitats to hibernate. The lack of winter dwellings is due to the modern humans' desire for clean landscapes, leading many property owners to tidy up their lawns and rake up and dispose of all the leaves. Environmentalists urge to keep the leaves available not only for hibernating animals (Yorkshire Wildlife Trust, 2024), but also for nutrient cycling and water absorption (Wohlleben, 2017). An interesting social media post referring to a mathematical experiment suggesting that the falling of leaves in autumn speeds up the earth rotation and makes our days shorter by 0.1 nanoseconds (Auburn.edu, 2019) was another inspiration to investigate this abundant resource.

To address the complex issues of cultural human expression and natural cycles in a way that would reconnect people to childhood play in a heap of leaves and their rustling sound, we filled the HSH with a pile of leaves that we needed to collect from one of our gardens. In this process we realised that the ones on Campus had already been removed as a case in point. Our autumn leaves intervention was shared in particular with the audience of the Bureau of Poetry Revisited Symposium at Stockholm School of Economics, exploring Poetic Imagination with Fire, Water, Air and Earth and the Art in Designed Living Environments Final Conference at ArkDes in Stockholm.

As visitors were stepping onto the forest floor, the leaves began to compress and break down into smaller pieces whilst the HSH was visiting Stockholm in contexts that explored designed living environment and the generativity of dreaming with the four elements fire, earth, air and water. Since we had some extra bags with leaves with us, we gradually added more, until the experience came close to moving through a dense little hill of leaves. There was an interesting moment of friction of how to best express seasonal designing in which one of us intended to fluff up the leaves, causing a reaction in another team member who thought: "*No, it's not about fluffing up the leaves, it´s about causing, witnessing, and acknowledging change and its different aesthetic expressions".*



Figure 4: A pile of leaves in the HSH. Svenja Keune, 2024

LIVING WITH HEATED UNCERTAINTIES (WINTER 2025)

A two-part intervention was inspired by the imagination to ice-skate around the table in the HSH, the joy of it, the contrast to how floors in a usual household are supposed to behave and the link to climate change and the vast effects that just 3 degrees difference can have. This intervention was set up during the Växjö Culture night and a few weeks later at Linnaeus University Campus in Växjö.

The THoW was sealed with a layer of plastic foil to prevent water damage to the wooden floor, then parked, leveled, and finally filled with a centimeter of water, covering the entire floor of the house. We had some rubber boots and ice-skates available for ourselves and the visitors, not being able to determine how the weather would be on the particular night that was announced as the Växjö culture night.

The weather on January 25, 2025 was rainy and around 5°C. Set up on a parking lot with lots of puddles, we also had a puddle inside the house and invited visitors to join us in this experience. In the darkness the puddle

was hardly visible and we had to “warn” our visitors when stepping in. Most shoe soles however were high enough for this level of water. We noted a change of behaviour when being in the puddle-dominated space. Walking became slower and more conscious, with an additional awareness to not drop anything on the floor.

Closely observing the weather forecast in the following weeks, an opportunity for a frozen floor opened up around the weekend of the 17th and 18th February. With a basic temperature of about -6°C we planned to flood the THoW again and invite to an indoor ice-skating experience. This frozen floor behaved very differently and cracked with the pressure of every step and skate.



Figure 5: The HSH on January 25 in Växjö. Svenja Keune, 2025

REFLECTIONS

The HSH as a physical and conceptual framework for exploring householding with resources maintained close relationships with the four seasons of the temperate climate of southern Sweden. Over time we engaged in closer relationships with seasonality. It became an environmental condition to enable transformation and to invite people to physically experience them, while reflecting about societal and environmental challenges of our time.

The unpredictability of which experience it is going to be brings to our attention the unpredictability of climate change and allows us to physically experience the difference that just 2-3 degrees can make in the experience of an environment and the consequential behaviours and actions that are possible or not. While we prefer plan a, a frozen floor, how can we deal with plan b, a puddle and its fundamentally driven implications on our ways of living and interacting with a space?

Here, aesthetics plays a major role, but it is not about addressing just the visual sense or freezing an impression but allowing for it to change and be experienced through multiple senses. Additionally, it is not a clean, lush, modern surface that is the aesthetical ideal, but a holistic aesthetical experience of life and death in tune with the seasons.

SEASONAL DESIGNING FRAMEWORK

After providing a series of seasonal interventions as examples of seasonal designing, this section introduces the Seasonal Designing Framework which is here outlined as compiled of six different aspects.

SEASONAL DESIGNING

Seasonal Designing is a design strategy that aligns with the rhythms of nature by working with material resources that are locally and seasonally available. Rather than treating materials as static, this approach acknowledges their capacity to change over time — degrading, drying, softening, or hardening in response to environmental conditions such as humidity, temperature, time, and light (Keune, 2018, 2019). Designing within this framework does not seek permanence but rather engages with transformation, embracing the ephemeral qualities of materials as part of their aesthetic and functional expression. By aligning not only the design process with seasonal cycles, designers tap into the surplus and scarcity inherent in local ecosystems, creating objects and structures that resonate with their temporal and geographical contexts.

SEASONAL RESEARCH APPROACH

The seasonal research approach situates a design process within a seasonal context and aligns with it (Keune, 2019). Similar to a gardener, who performs different activities at different times of year, the type of research activities and the timing of events are determined by the seasons, i.e. temperature, weather, amount of light, activity of the lifes of plants, animals, and fungi. It thereby implies a letting go of the dominant attempts at controlling nature through culture.

SITUATED ENVIRONMENT

Instead of using a controlled environment, this approach considers the natural environment and ecosystem in which the organisms operate. The seasons and the situated environment give cues for design efforts. The seasons become an active co-creator and environment in which to conduct an experiment or place/expose an artefact or installation (Seasonal Intervention).

MATERIALS FOR DESIGNING

Elements such as water, fire, earth, and wind and seasonal expressions such as hail, snow, ice, rain, sunshine/UV radiation, and seasonal expressions of plants and landscapes become resources and determine the materials for seasonal designing.

AESTHETIC LANGUAGE

The aesthetic language is determined by the seasonal expressions of the materials and is temporary.

Expressions of decay and natural colours are dominating.

SEASONAL INTERVENTION

Seasonal Interventions operate within the dynamic interplay of weather, animals, plants, and human activity, treating the environment itself as a resource and collaborator. These interventions do not extract but rather interact, responding to shifting conditions—rain, wind, migration patterns, or community rhythms. Such practices might involve water in its different states, using wind to shape fabric, or inviting birds and insects into designed structures (Keune & Lim, 2024). Here, design is not imposed onto the landscape but emerges through attunement to its forces, fostering social, economic and ecological relationships that are reciprocal rather than extractive.

CONCLUDING DISCUSSION

Engaging with the seasons—i.e., the yearly changes that mark periods in southern Sweden, including weather and shifts in light, humidity, temperature, atmosphere, and perception of time—provided an accessible entry point to understanding and discussing resources beyond their material level. Instead, resources emerged as a situated and temporal force that we can work with and relate to.

Our seasonal experiments have taken place in streets, museums, formal educational settings, and informal learning environments, supporting lifelong learning. The seasonal designing framework thus has the potential to travel to wherever designers and researchers want to anchor it.

SEASONAL DESIGNING AS ACTIVISTIC ACT

Seasonal changes are increasingly affected by climate change, causing challenges for many ecosystems, their co-creators, and inhabitants. Linking design processes with the seasons provides a way to witness and deeply understand these relational dynamics, enabling design-driven responses to environmental shifts.

Seasons also significantly impact human and more-than-human sensory experiences and mental, emotional, and physical well-being. Historically, this has led to technological innovations driven by capitalist values. Rethinking human and more-than-human relationships with seasons is thus a critical design objective in supporting a transition toward more regenerative and sustainable practices in both design and society.

SEASONS AS MATERIAL FOR DESIGN AND AS AESTHETICAL LANGUAGE

Embracing the expressions of e.g. growth and decay as part of design process. This is a major turn away from controlled design expressions that has the same appearance over time. In addition, it is a move away from desires of the new and the shiny. Embracing the natural expressions of growth and decay within the design process marks a major departure from controlled design aesthetics that maintain a static appearance over time. It also challenges conventional desires for the new and the pristine, encouraging a design approach that acknowledges temporal change as part of its aesthetic language.

ENVIRONMENTAL CHALLENGES APPROACHED THROUGH THE SEASONS THEY ARE SITUATED IN

Environmental challenges can be more meaningfully engaged with when understood through the seasons in which they manifest. For example, the *Living with a Forest Floor* intervention in autumn addresses the lack of habitat for hibernation. Working with seasons fosters a deeper connection with nature and highlights the impacts of environmental shifts on human and more-than-human life.

SEASONS AS ENVIRONMENTAL CONDITION AND CO-CREATOR FOR DESIGN.

Autumn was the environmental condition and framework for the *Living with a Forest Floor* intervention to address the lack of habitat for hibernation that is caused by cleaning up leaves. During the intervention the leaves also became the co-creators. The leaves changed the visual, auditory, and olfactory expression of the interior of the HSH. Such interventions are both locally anchored and globally relevant.

Seasonal expressions inevitably find their way into domestic spaces, as seen in Keune's Textile Farming research and the concept of seasonal interiors (Keune, 2019) or in the *+Change* education programs at Linnaeus University (Ståhl, 2023, 2024). Householding with resources becomes a socio-material commitment, requiring an ongoing negotiation of relationships as a strategy for becoming-with others.

Designing with seasons entails scaling in multiple ways —geographically, size-wise, and across permeable boundaries such as walls or skin. This approach acknowledges the frictions and oppositions that arise when different householders—whether human or more-than-human—co-live. Rather than seeking seamlessness and harmony as the only outcome, seasonal designing embraces generative friction and alternative ways of living together, enabling transformative and liveable futures.

TOWARD A SEASONAL APPROACH TO DESIGN

As a contribution to design students and practitioners, this paper encourages those conducting design projects to:

a) Align their design work with the seasons as an accessible starting point to relate to the environment, local context, and the more-than-human.

b) Stage interventions that enable designers and others to experience and explore environmental challenges in playful ways, fostering creative and innovative everyday practices that can scale beyond the initial situation.

Ultimately, householders—whether individuals, communities, or entire ecosystems—are diverse yet must co-live. Design credos of seamlessness and harmony are not to be expected as the only result of designing with seasons. Instead, seasonal designing offers a framework for embracing difference, fostering resilience, and navigating the complexities of a shared world.

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